OPERATING SYSTEMS

Process Management and Scheduling

S.Thenmozhi

Department of Computer Applications

- Process Scheduler assigns the process to the CPU following a Scheduling Algorithm
- Six scheduling algorithms widely used
  - First-Come, First-Served (FCFS) Scheduling
  - Shortest-Job-First (SJF) Scheduling
  - Shortest Remaining Time
  - Priority Scheduling
  - Round Robin(RR) Scheduling
  - Multiple-Level Queues Scheduling
- Algorithms are either non-preemptive or preemptive

- Non-preemptive algorithms are designed so that once a process enters the running state, it cannot be preempted until it completes its allotted time
- Preemptive scheduling is where a scheduler may preempt a low priority running process anytime when a high priority process enters into a ready state.

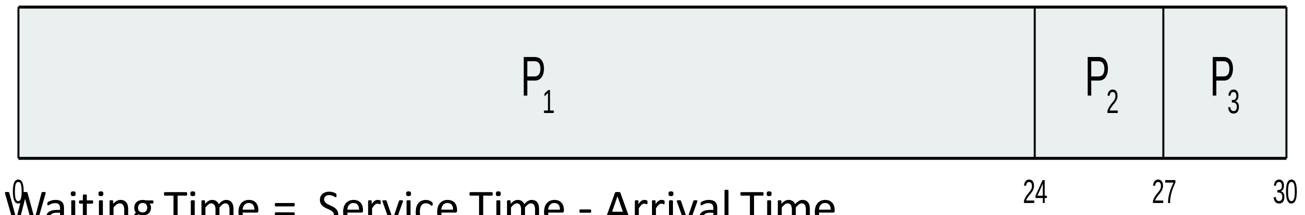
FCFS – First Come First Served

- Jobs are executed on first come, first serve basis
- It is a non-preemptive scheduling algorithm
- Easy to understand and implement
- Its implementation is based on FIFO queue
- Poor in performance as average wait time is high

FCFS - First Come First Served

Process	Burst Time
P1	24
P2	3
P3	3

- Suppose that the processes arrive in the order: P_1 , P_2 , P_3
- The Gantt Chart for the schedule is:



- Waiting Time = Service Time - Arrival Time
- Waiting time for P_1 = 0; P_2 = 24; P_3 = 27
- Average waiting time: (0 + 24 + 27)/3 = 17

FCFS - First Come First Served

Suppose that the processes arrive in the order: P_2 , P_3 , P_1

- The Gantt chart for the schedule is:

- Waiting time for $P_1 = 6; P_2 = 0; P_3 = 3$
- Average waiting time: $(6 + 0 + 3)/3 = 3$
- Much better than previous case
  - Hence, average waiting time of FCFS not minimal
  - And it may vary substantially
- FCFS is nonpreemptive
- Not a good idea for timesharing systems
- FCFS suffers from the convoy effect, shorter job to wait for a longer time

SJF – Shortest Job First

- This is also known as **Shortest Job Next**, or SJN
- This is a non-preemptive, pre-emptive scheduling algorithm.
- Best approach to minimize waiting time.
- Easy to implement in Batch systems where required CPU time is known in advance.
- Impossible to implement in interactive systems where required CPU time is not known.
- The processer should know in advance how much time process will take.

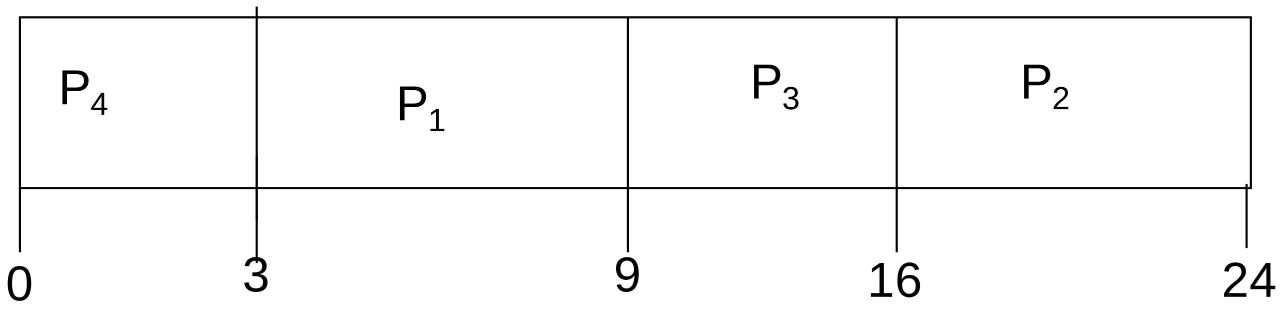
SJF – Shortest Job First

- Assume
  - A new process Pnew arrives while the current one Pcur is still executing
  - The burst of Pnew is less than what is left of Pcur
- Nonpreemptive SJF – will let Pcur finish
- Preemptive SJF – will preempt Pcur and let Pnew execute

SJF – Shortest Job First

Process	Burst Time
P1	6
P2	8
P3	7
P4	3

- SJF scheduling chart



- Average waiting time = (3 + 16 + 9 + 0) / 4 = 7

THANK YOU

S. Thenmozhi
Department of Computer Applications
thenmozhis@pes.edu
+91 80 6666 3333 Extn 393